

Abhishek Tagalpallewar

Junior Undergraduate
Mechanical Engineering, Minors in Computer Science and Engineering
Indian Institute of Technology Gandhinagar

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ACADEMIC DETAILS

Degree	Institution	Year	CPI/%
B.Tech	Indian Institute of Technology Gandhinagar	2024-2028	8.70
Class XII	Late P. B. Jog Junior College (MSBSHSE)	2024	89.5
Class X	Mahesh Vidyalaya English Medium School	2022	95.8

PROJECTS

Neural Network Framework for Learning Constitutive Laws in 2D Granular Flow [Jan'26-Apr'26]
(Advisor - Prof. Harkirat Singh, IIT Gandhinagar) | Project Link

- Engineered modular **PyTorch** framework evaluating **3** stress components (sigma-xx, sigma-yy, sigma-xy) across **25** settings
- Reduced prediction error rates from **48%** to **3%** using reference normalization models on **2** unseen datasets
- Automated **2D spatial grid mapping** to handle large-scale internal **2D stress tensor** variations in training

Four-Bar Mechanism Kinematic Analysis [Aug'25-Nov'25]
(Advisor - Prof. K. R. Jayaprakash, IIT Gandhinagar) | Project Link

- Developed a **MATLAB** simulation engine for planar 4-bar linkages using vector-loop kinematics to compute link motion profiles
- Programmed computer vision algorithms in **Python** using high-speed **211.55 FPS** video tracking to verify analytical loops
- Optimized rigid body kinetics by scripting data pipelines to resolve relative velocity (V-G) and angular velocity (Omega)

Validation of Angular Momentum Conservation via Reynolds Transport Theorem [Jan'26-Apr'26]
(Advisor - Prof. Shubhangi Bansude, IIT Gandhinagar) | Project Link

- Validated physical milestones by fabricating a custom **3-nozzle** rotating water sprinkler prototype
- Formulated a predictive **MATLAB** framework to model relative error variances of **31.1%** and **42.0%** across **2** speeds

RELEVANT COURSEWORK

- Thermodynamics | Fluid Dynamics | Mechanics of Solids | Statics & Dynamics | Manufacturing Processes | Vibrations

TECHNICAL SKILLS

- Programming Languages:** Python | C | C++
- Developer Tools:** VS Code | Git | GitHub
- Libraries:** NumPy | Pandas | Matplotlib | PyTorch
- Software & Simulation:** MATLAB | Cantera | Mathematica | Autodesk Inventor | MSC Adams (Basic) | ANSYS Fluent (Basic) | Creality Print

ACHIEVEMENTS

- Secured **Dean's List** (top 5%) and **Rank 1 (Branch Topper)** in Mechanical Engineering with **9.48 SPI** in Semester 3
- Earned **A+ (11/10)** in **Thermodynamics** for exceptional performance across advanced curriculum evaluations
- Won Division **4** with the chess team at India Chess Championship 2025 (**250+** teams across **30** divisions)
- Claimed **Team Runner-up** at DA-IICT Chess Tournament and individual runner-up honors at IIT Bombay Sports Fest

POSITIONS OF RESPONSIBILITY

- Coordinating Team Member | Institute Sports Events, IIT Gandhinagar** [Jul'24-Apr'26]
 - Managed basic operational logistics, scheduling, and tracker loops for **3+** major sports festivals
 - Collaborated closely with a **15-member** student council to streamline tournament timelines
- Junior Team Member | Amalthea & Blithchron Festivals, IIT Gandhinagar** [Aug'24-Feb'25]
 - Coordinated core event operations during Amalthea'25 by assisting with hospitality logistics
 - Tracked company outreach and record matrices to maintain partner communications for Blithchron'25

EXTRA-CURRICULAR ACTIVITIES

- Represented IIT Gandhinagar twice at the **Inter IIT Sports Meet**, competing in Chess matches against **20+** IITs
- Participated in chess tournaments and inter-university matches hosted at Ahmedabad University and IIT Jodhpur